Military Specification
Cannon, 105MM Howitzer: M2A2

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of howitzer cannon.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannon, General Specification for

MIL-P-14232 - Parts, Tools and Equipment for Army Material, Packaging of

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance, and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-LCB-SAS, Watervliet, N.Y. 12189-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1015

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STANDARDS

Military

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-1189 - Standard Symbology for Marking Unit Packs, Outer Containers, and Selected Documents

MIL-STD-1235 - Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. Army Armament, Munitions and Chemical Command

QAP-APPENDIX-WVA - General Quality Assurance Provisions

EPL7827208 - Engineering Parts Lists or Cannon, 105MM Howitzer: M2A2

LSPI7827208 - List of Special Packaging Instructions, Cannon, 105MM Howitzer: M2A2

Master LQAP 7827208 - Master Index of Quality Assurance Provisions for Cannon, 105MM Howitzer: M2A2

DL7827208 - Index of Inspection Equipment Lists for Cannon, 105MM Howitzer: M2A2

P7827208 - Special Packaging Instructions Cannon, 105MM Howitzer: M2A2

PUBLICATION

DA Form 2408-4 - Weapon Record Data

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 -Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, drawing requirements shall have first order of precedence, followed by the text of this specification, then the other referenced specifications and standards.

3. REQUIREMENTS

3.1 First article. The contractor shall submit a first article unless it is specifically waived in the contract (see 6.2). No first article requirements shall be waived without review by the responsible engineering activity and approval by the responsible product assurance activity.

3.2 General requirements. Cannon and parts thereof shall conform to the drawings listed in EPL 7827208, to the requirements of this specification, and to MIL-C-13931.

3.3 Functional characteristics. Each cannon assembly shall perform the firing and pressure sustaining function of the 105MM: M2A2 Howitzer. All moving parts shall function smoothly without interference, erratic movement or malfunction.

3.3.1 Breech mechanism. When the operating lever is unlatched and rotated rearward (clockwise), it shall cam the breechblock horizontally to the right (facing the breech) and thereby open the breech. After the insertion of a round in the chamber, the breechblock shall be cammed back into the breech ring by rotating the operating lever forward (counterclockwise). During this closing movement, the beveled (front) face of the breechblock shall contact the base of the cartridge case and force it to seat in the chamber. When the breech is fully closed, the operating lever shall be locked in closed position by the catch. During the closing of the breech, the extractor path in the bottom of the breechblock shall cam and rotate the extractor until its lip is positioned in front of the rim of the cartridge case. Then, the opening of the breech shall cam the extractor lip sharply rearward, thereby extracting and ejecting the case from the chamber.

3.3.2 Firing mechanism group. The firing mechanism consists of the Firing Lock M13 which is operated by movement of the trigger-shaft. For firing, the trigger-shaft-lever-arm is rotated clockwise (by pulling the lanyard outward). This action shall rotate the trigger-fork forward to force the sleeve forward and compress the firing-spring. The moving sleeve shall trip the sear which shall release the firing-pin-holder and allow the compressed firing-spring to expand and drive the firing-pin forward to strike the primer. The firing-spring shall then force the sleeve and firing pin holder rearward, which in turn shall rotate the trigger fork clockwise. The sleeve engages the sear and thereby completes the firing cycle.

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3.3.3 Firing pin protrusion. When fired, the protrusion of the firing pin from the front of the breechblock shall be not less than 0.164 inch nor more than 0.296 inch.

3.3.4 Headspace. Headspace, the distance from the front face of the closed and locked breechblock and the rear face of the tube, shall be not less than 0.189 inch nor more than 0.209 inch.

3.4 Performance.

3.4.1 High pressure resistance. The cannon and repair tubes shall be capable of withstanding a chamber pressure of 44,100 plus or minus 1,600 pounds per square inch when tested as specified in 4.6.6.1.

3.4.2 Material soundness. After testing, all parts shall be free of cracks and fractures and shall comply with any soundness requirements specified on the applicable drawings.

3.4.3 Bore enlargement. If proof firing is used to evaluate high pressure resistance, bore diameters (measured across the lands) after proof firing no more than 10 rounds shall not exceed the before firing diameters by more than the amounts specified in TABLE I.

TABLE I. Bore enlargement.

Distance forward of commencement of rifling	0.1 inch	1 inch	1 cal. to muzzle
Amount (in inches)	0.003	0.003	0.002

NOTE: Interpolate to determine the amount between given distance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Governemnt reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The inspection set forth in the specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of his responsibility for assuring that all supplies and services submitted to the Government for acceptance conform to all requirements of the contract.

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4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. First Article - Initial Production Test, FA-IPT (see 4.5).

c. Quality conformance inspection (see 4.6).

4.4 First article inspection. Unless otherwise specified in the contract, (see 6.2.1), First Article cannon and specified parts thereof shall be inspected by the contractor and submitted to the Government for approval. Inspection shall be in accordance with applicable Quality Assurance Provisions (QAPs) and QAP-APPENDIX-WVA. QAPs are listed in Master LQAP 7827208. First Article sample sizes shall be as specified in applicable item QAPs, or the contract, as applicable. First Article cannon and parts thereof shall be representative of the production processes to be used during quantity production.

4.5 First Article - Initial Production Test (FA-IPT). When specified in the contract, (see 6.2.1), First Article cannon shall be subjected to Initial Production Testing by the Government or any agency specified by the Government and may be conducted at a Government facility to assure compliance with the requirements of this specification and any additional requirements listed in the contract. Initial Production Testing procedures and the sample size of first article cannon shall be as specified in the contract. First article cannon and parts thereof shall be representative of the production processes to be used during quantity production.

4.6 Quality conformance inspection. Quality conformance inspection is listed in TABLE II.

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TABLE II. Quality conformance inspection.

Category	Requirement Paragraph	Inspection Paragraph	Inspection method Paragraph
Examinations			
Firing pin protrusion	3.3.3	4.6.4.1	4.6.4.1
Headspace	3.3.4	4.6.4.2	4.6.4.2
Material soundness	3.4.2	4.6.4.3	4.6.4.3
Packaging	5.	4.6.4.4	4.6.4.4
Functional inspection	3.3	4.6.5	4.6.5
Breech mechanism	3.3.1	4.6.5.1	4.6.5.1
Extractors	3.3.1	4.6.5.2	4.6.5.2
Firing mechanism group	3.3.2	4.6.5.3	4.6.5.3
Tests			
Proof acceptance	3.4.1	4.6.6.1	4.7.1
	3.4.2		4.7.2
	3.4.3		

4.6.1 Inspection provisions. Cannon and parts thereof shall be inspected for conformance to requirements of this specification, MIL-C-13931, QAPs listed in Master LQAP 7827208, and QAP-APPENDIX-WVA.

4.6.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.6.3 Inspection equipment. Data lists containing equipment used in the inspection of parts, subassemblies, and assemblies are listed in DL 7827208.

4.6.3.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.2.1), responsibility for acquisition, maintenance, and disposition of acceptance inspection and test equipment referenced in the List of Inspection Equipment DL 7827208, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

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4.6.3.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6.4 Examinations.

4.6.4.1 Firing pin protrusion. Firing pin protrusion of the breech mechanism of each cannon shall be measured using gage 5256336 or approved equivalent to determine compliance with 3.3.3.

4.6.4.2 Headspace. Headspace, as defined in 3.3.4, shall be measured on each cannon using gages 8775727 and 8775728 or approved equivalent gages to determine compliance with 3.3.4.

4.6.4.3. Material soundness. All external surfaces of the tube and all surfaces of the breech ring and the breechblock shall be examined by magnetic particle inspection in accordance with applicable drawings and specifications. Acceptance criteria shall be as stated on the applicable drawings. The bore surfaces of the tube shall be examined by means of a borescope. There shall be no evidence of cracks or fractures.

4.6.4.4 Packaging inspection. Packaging inspection for the level designated in the contract shall be performed in accordance with the applicable packaging data sheets and MIL-P-14232. Packaging failing to comply with the packaging requirements of Section 5 and the contract shall be cause for rejection.

4.6.5 Functional inspection. Functioning tests shall be conducted in accordance with MIL-C-13931 and as specified herein. A minimum of two complete cycles of operation shall be performed on each cannon.

4.6.5.1 Breech mechanism. Each breech mechanism shall be operated manually to determine smoothness of operation in compliance with 3.3.1. Action of the operating lever shall be observed to assure that the lever is locked when in the closed position. Any evidence of interference, erratic movement, or malfunction of any component shall be cause for rejection.

4.6.5.2 Extractors. With the cannon in a horizontal position, extraction and ejection shall be tested with an empty service cartridge case to determine compliance with 3.3.1.

4.6.5.3 Firing mechanism group. When assembled in a breechblock or breech mechanism, each firing mechanism shall be operated manually to determine smoothness of operation in compliance with 3.3.2. Any evidence of interference, erratic movement, or malfunction of any component shall be cause for rejection.

4.6.6 Tests.

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4.6.6.1 Proof acceptance. Cannon shall be proof accepted through proof firing, proof pressure simulation tests, or both. Cannon and repair tubes to be proof fired shall be fired using the test method of 4.7.1. Failure to comply with the requirements of 3.4.1, 3.4.2, and 3.4.3 shall be cause for rejection. Cannon breech mechanisms to be simulation proof tested shall be proof tested using the test method of 4.7.2. Failure to comply with the requirements of 3.4.1 and 3.4.2 shall be cause for rejection. Proof firing performed on a sampling basis shall be conducted as follows:

a. Using plan CSP-1 of MIL-STD-1235 and an AQL of 1.0, the Product Assurance Directorate at Watervliet Arsenal shall designate the appropriate initial quantity (i) and sample frequency (f), in the contract (see 6.2.1).

b. The first "i" quantity of cannon or tubes produced in a particular buy shall be proof fired. When the "i" quantity has been satisfied, sampling shall be put into effect using the "f" sampling frequency. All cannon not proof fired shall have the breech mechanisms simulation proof tested in the Product Assurance weapon test facility at Watervliet Arsenal.

c. Successive production buys shall not require "i" quantities. However, if the time lag between two successive buys exceeds one year, or if a new cannon or tube contractor is utilized, a new "i" quantity shall be proof fired.

d. Proof fired cannon and replacement tubes, and simulation proof tested breech mechanisms shall be furnished with applicable data on DA Form 2408-4 (Weapon Record Data) as specified by Product Assurance Directorate at Watervliet Arsenal.

e. When specified in the contract (see 6.2.1), the weapon simulation test facility shall be qualified as follows: a sample of two breech mechanisms of cannon to be sent to the proving ground for proof firing shall be subjected initially to simulation proof testing at the weapon simulation test facility.

4.7 Test methods.

4.7.1 Proof firing. Proof firing shall be conducted at a Government test site. The procedure shall be as follows:

a. Each cannon or repair tube shall be proof fired once within the proof pressure range specified in 3.4.1.

b. A facility mount may be used if necessary.

c. Chamber pressure shall be measured and recorded.

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d. After proof firing:

(1) The bore of each tube shall be examined for bore enlargement using an air gage or other approved gage. Enlargements exceeding the limits specified in TABLE I shall be cause for rejection (see 3.4.3).

(2) Each tube shall be borescoped and magnetic particle inspected (see 4.6.4.3).

(3) Each breech mechanism, including the extractors, shall be examined for looseness or interference of fit and other defects of workmanship and function (see 4.6.5.1 and 4.6.5.2).

(4) Each firing mechanism group shall be examined for looseness or interference of fit and other defects of workmanship and function (see 4.6.5.3).

(5) Each breech mechanism shall be disassembled and examined for soundness of material. All components shall be examined to assure freedom from cracks and fractures. Each breech ring and breechblock shall be magnetic particle inspected (see 4.6.4.3).

e. Cannon found to be defective at the proving ground shall be reported to the Product Assurance Directorate, Watervliet Arsenal, Watervliet, NY 12189-5000 and Benet Weapons Lab, Watervliet, NY 12189-5000 within one working day. The test agency shall forward an Equipment Preformance Report (EPR), incident to the deficiency, within three working days.

4.7.2 Proof pressure simulation. All cannon not proof fired at a proving ground shall have the breech mechanism proof tested in accordance with the following procedure:

a. Each breech mechanism shall be tested by means of suitable hydraulic pressure testing equipment at a pressure within the proof pressure range specified in 3.4.1.

b. Cycle time to achieve test pressure and release shall not exceed three minutes. Test pressure shall be immediately released upon its attainment. Dwell time at test pressure shall not exceed three seconds.

(1) When difficulty is encountered in attaining test pressure, the number of attempts in which 10 percent or less of the test pressure is attained shall not be counted. The number of attempts in which over 10 percent of the test pressure is attained shall not exceed five.

(2) If the cause of difficulty is not identified and corrected within five pressure cycles, the breech mechanism shall be subjected to proof firing as part of a cannon (see 4.7.1). A notification of this condition shall be submitted with the breech mechanism to the proving ground.

c. The peak pressure shall be recorded for each test and attached to inspection records identifiable with the specific breech mechanism undergoing test.

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d. After the pressure test, each breech mechanism shall be examined for looseness or interference of fit and other defects of workmanship and function (see 4.6.5.1).

e. Each breech mechanism shall be disassembled and examined for soundness of material. All components shall be visually examined to assure freedom from cracks and fractures. Each breech ring and breechblock shall be magnetic particle inspected (see 4.6.4.3).

5. PACKAGING

5.1 Cannon - Levels A, B and C. Preservation, packaging and packing of the cannon shall be in accordance with Special Packaging Instructions 7827208 for Cannon, 105MM Howitzer M2A2 for the level of protection specified in the contract (see 6.2.1).

5.2 Repair parts.

5.2.1 Levels A, B, and C. Repair parts preservation, packaging and packing shall be in accordance with the applicable Special Packaging Instructions listed in the list of Special Packaging Instructions LSPI 7827208 for the level of protection specified in the contract (see 6.2.1).

5.2.2 Level X-industrial packaging. Repair parts preservation, packaging, and packing shall be in accordance with ASTM D 3951 when Level X protection is specified in the contract (see 6.2.1).

5.3 Marking.

5.3.1 Levels A, B and C. Marking of unit packages and exterior shipping containers shall be in accordance with Packaging Data Sheet 7827208 for Cannon, 105MM Howitzer M2A2 for the level of protection specified in the contract (see 6.2.1).

5.3.2 Level X - Industrial. Marking shall be in accordance with ASTM D 3951.

5.3.3 Bar code marking. When specified, (see 6.2.1.n) bar code marking shall be in accordance MIL-STD-129 and MIL-STD-1189.

6. NOTES

6.1 Intended use. The 105MM M2A2 is intended for use in the M101 and M101A1 Howitzer systems.

6.2 Ordering data. Procurement documents should specify the following:

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6.2.1 Acquisition requirements.

a. Title, number and date of this specification.

b. Requirements for and the number of cannon, components thereof, or both to be submitted for first article by the contractor (see 3.1).

c. The examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

d. All inspection and test data that is required.

e. At whose expense a retest may be performed.

f. Serial numbers for the cannon and replacement tube assemblies.

g. Inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.6.3.1).

h. Availability of inspection equipment from the Government.

i. Extent of supplier's responsibility for Government furnished and for supplier-required final inspection equipment

j. Applicable acceptance test procedures, initial consecutive quantity (i), and sampling frequency (f) (see 4.6.6.1).

k. Under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

l. When applicable, the sampling plan for the actual firing of replacement tubes.

m. The levels of preservation, packaging, and packing required for cannon assemblies (see 5.1) and repair parts (see 5.2).

n. When applicable, bar code marking requirements (see 5.3).

6.2.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be as specified for delivery on DD Form 1423 in the contract.

6.2.2.1 When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD FAR Supplement 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

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Paragraph No.	Data Requirement Title	Applicable DID No.
3.1	First Article Inspection Report	DI-T-4902

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Volume II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Contracting Officer).

6.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate.

6.4 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-STD-45662.

6.5 Definition. Frequency of testing is defined as the proportion of the lot to be tested after 100 percent testing.

6.6 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreements, QSTAG-150. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

Custodian:

Army-AR

Preparing activity:

Army-AR

Project No. 1015-A314

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-14384C	CANNON, 105MM HOWITZER: M2A2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION (YYMMDD)

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